

Ice Core

Earth & environmental science

A deep-drilling camp on a polar plateau, with the flattest, emptiest horizon in the set.

45	14	6.1	9/10	2	19·24·2
STOPS	CHOICE (31%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT·CONSTRUCT·OPERATE

READING Where this campaign stands

Strong: 9 of 10 computed and the remaining gap sits on a BALLPARK that looks like it already does the arithmetic, so Phase 0 will probably clear it to zero. Two select-only concepts and two absent ones. Fourteen CHOICE stops is the only real weakness, and the place is unusually good for the formats that would replace them: a core line, a cold laboratory, a drill trench under a tower, and flag lines across a plateau.

NOW The mix as it ships

CHOICE 14	BALLPARK 10	PROTOCOL 3	SEQUENCE 3	DIAGNOSIS 2	TRIAGE 2	SWEEP 1	VALUE 1	CLOUD 1	CHAIN 1	SCIENCETANK 1	PROPAGATE 1	TRACE 1	VERIFY 1	ALLOCATE 1	ATTEST 1	STRESS 1
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CURRICULUM What the syllabus is not getting

EQUATION	THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$N(t) = N_0 e^{(-t/\tau)}$		radiometric dating, where nothing can be counted	32·BALLPARK	real gap

0 of 1 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 1 is unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
17	Sea salt and the chemistry that reaches the ice		6, 8, 20
23	Core logging, orientation and chain of custody		5

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 2 concepts no stop touches at all — **Cosmogenic ¹⁰Be, solar activity and geomagnetic markers; Splicing a proxy record onto the instrumental one** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

PROBE

STRONG

stop 5 or 8 · CORE / COLD

"The metre with no label" is a CHOICE. Take readings down the core, depth by depth, and find where layer counting stops working — which is not the depth with the worst number on it. The core line is a chain in the most literal sense in the catalogue.

SERVES concept 23 — core logging and chain of custody — and depth-age dating

NEEDS a profile with one honest break

TALLY STRONG

stop 8 or 32 · COLD / CORE

Counting annual layers until the date is defensible **is** ice-core dating, and it is the same decision TALLY grades: when is there enough behind the number to report it.

SERVES layer counting against radiometric dating, $N(t) = N_0 e^{(-t/\tau)}$

NEEDS an early count that flatters the date

BELT STRONG

CORE · the processing line

A core-processing line is a conveyor with a sort on it, and the one bit of subject matter is real: this section is for gas, that one for isotopes, and a mis-sorted section cannot be recovered. Concept 23 is select-only.

SERVES concept 23, chain of custody, at the speed a line moves

RISK must be decidable at a glance

HOLD STRONG

DRILL · the trench

Hold borehole fluid pressure inside a closing band as the hole deepens. Get it wrong and the hole closes — a consequence that is physical, expensive and true.

SERVES pressure with depth, and drilling as a controlled rate

TRIAL STRONG

the flag lines

Flag lines exist on this site because in a whiteout they are the only way to know where you are. A driven route between modules, graded on order — and the format's own sentence is "learn it as places, not as a count of paces."

SERVES field discipline; the flattest, emptiest horizon in the set made load-bearing

RISK world work

SWEEP MEDIUM

a GAS or DATA stop

A proxy against a driver, plotted only where you look. The game already carries one SWEEP, so a second competes with itself.

SPOT MEDIUM

FIELD · Snow Study

A sampling instruction revised mid-traverse. Fits; medium because the game's ATTEST already argues about records.

LOB REJECT

Reject on subject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs icecore --snapshot before.json, then --against before.json.

Minimum honest 3 STOPS · ONE SITTING

5	d2	CORE	CHOICE → PROBE	The metre with no label the metre with no label, read down the core
8	d3	COLD	CHOICE → TALLY	Which year marker survives the depth layers counted until the marker is defensible
20	d7	CORE	CHOICE → SWEEP	Four times the dust four times the dust — as a curve against its driver

CHOICE **14** → **11** (24%) · moves **19·24·2** → **16·24·5** · effective formats **6.1** → **7.7** · **3** new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

5	d2	CORE	CHOICE → PROBE	The metre with no label
8	d3	COLD	CHOICE → TALLY	Which year marker survives the depth
20	d7	CORE	CHOICE → SWEEP	Four times the dust
6	d2	COLD	DIAGNOSIS → BELT	The sample that was clean until it was handled the processing line, sorted before it warms
32	d11	CORE	BALLPARK → HOLD	The clock that does not need layers the clock that does not need layers — held against decay
43	d15	DATA	CHOICE → SPOT	The sentence the record will carry a sampling instruction revised mid-traverse

CHOICE **14 → 10** (22%) · moves **19·24·2 → 14·23·8** · effective formats **6.1 → 9.1** · **6** new operate stops, 3 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

5	d2	CORE	CHOICE → PROBE	The metre with no label
8	d3	COLD	CHOICE → TALLY	Which year marker survives the depth
20	d7	CORE	CHOICE → SWEEP	Four times the dust
6	d2	COLD	DIAGNOSIS → BELT	The sample that was clean until it was handled
32	d11	CORE	BALLPARK → HOLD	The clock that does not need layers
43	d15	DATA	CHOICE → SPOT	The sentence the record will carry
12	d4	GAS	PROTOCOL → TRIAL	Which differences a shared instrument removes the flag lines, driven, in a whiteout
27	d9	CORE	CHOICE → RESIDUAL	The last hundred metres what the proxy fit leaves over
38	d13	DRILL	TRIAGE → INJECT	What comes back first a known signal pushed through the game's own pipeline

CHOICE **14 → 9** (20%) · moves **19·24·2 → 12·24·9** · effective formats **6.1 → 10.2** · **7** new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline `curriculum-debt.json`. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — `tools/import-book.mjs` `books/icecore.yml` `icecore --verify`. **4.** `npm run traps`: every instrument's trap must fire. **5.** `npm run check icecore` and `npm run drive icecore`, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** `--against` the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/icecore/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=icecore npm run dev`.